

Identify the points of discontinuity, holes, vertical asymptotes, x-intercepts, and horizontal asymptote of each.

$$1) f(x) = \frac{1}{3x^2 + 3x - 18} = \frac{1}{3(x-2)(x+3)}$$

$$HA: y=0$$

$$VA: x=2, -3$$

$$3) f(x) = \frac{x^3 - x^2 - 6x}{-3x^2 - 3x + 18}$$

slant asymptote

$$HA: y=3, -2$$

$$VA: x=-3, 2$$

$$2) f(x) = \frac{x-2}{x-4}$$

$$HA: y=2$$

$$VA: x=4$$

$$(x+3)(x-2)$$

$$-4(x+3)(x+1)$$

$$HA: y=-3, -2$$

$$VA: x=-3, -1$$

Find all asymptotes of the following functions. (Do not graph these)

$$A) f(x) = \frac{x-7}{x+5}$$

$$HA: y=7$$

$$VA: x=5$$

$$B) f(x) = \frac{3}{x^2-2}$$

$$HA: y=0$$

$$VA: x=\pm\sqrt{2}$$

$$C) f(x) = \frac{x^2}{x-5}$$

$$HA: y=0$$

$$VA: x=5$$

$$D) f(x) = \frac{2x^2 - 5x + 3}{x-1}$$

$$HA: y=\frac{3}{2}, 1$$

$$VA: x=1$$

$$E) f(x) = \frac{7x^2 + 5x - 2}{2x^2 - 18}$$

$$HA: y=-1, 2$$

$$VA: x=\pm 3$$

$$F) f(x) = \frac{2x^2 - 5x + 5}{x-2}$$

not factorable

$$VA: x=2$$

$$G) f(x) = \frac{1}{3-x}$$

$$HA: y=0$$

$$VA: x=3$$

$$H) f(x) = \frac{x^2-4}{x^4-81}$$

$$HA: y=\pm 2$$

$$VA: x=\pm 3$$

$$I) f(x) = \frac{x^3 - 2x^2 + 5}{x^2}$$

$$x^2 \begin{array}{|c|c|c|} \hline x^3 & -2x^2 & 5 \\ \hline \end{array}$$

$$x^3 - 2x^2 + 0x + 5$$

$$HA: y=-2$$

$$VA: x=0$$